

Homework 5

1 Problem 1 (B-5-29)

Consider the system defined by:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -25 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad (1)$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad (2)$$

To calculate the transfer matrix of this system, solve $(G(s) = C(sI_2 - A)^{-1} + D)$, where:

$$A = \begin{bmatrix} 0 & 1 \\ -25 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad C = D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad (3)$$

First, compute $(sI_2 - A)$:

$$sI_2 - A = \begin{bmatrix} s & -1 \\ 25 & s + 4 \end{bmatrix} \quad (4)$$

Next, compute the inverse of this matrix:

$$(sI_2 - A)^{-1} = \left(\frac{1}{s^2 + 4s + 25} \right) \begin{bmatrix} s + 4 & 1 \\ -25 & s \end{bmatrix} \quad (5)$$

Finally, we can compute the transfer matrix of the system:

$$G(s) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \left(\left(\frac{1}{s^2 + 4s + 25} \right) \begin{bmatrix} s + 4 & 1 \\ -25 & s \end{bmatrix} \right) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (6)$$

$$= \left(\frac{1}{s^2 + 4s + 25} \right) \begin{bmatrix} s + 4 & 1 \\ -25 & s \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (7)$$

$$= \left(\frac{1}{s^2 + 4s + 25} \right) \begin{bmatrix} s + 4 & s + 5 \\ -25 & s - 25 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (8)$$

$$= \begin{bmatrix} \frac{s+4}{s^2+4s+25} & \frac{s+5}{s^2+4s+25} \\ -\frac{25}{s^2+4s+25} & \frac{s-25}{s^2+4s+25} \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (9)$$

$$= \begin{bmatrix} \frac{(s+4)+(s^2+4s+25)}{s^2+4s+25} & \frac{s+5}{s^2+4s+25} \\ -\frac{25}{s^2+4s+25} & \frac{(s-25)+(s^2+4s+25)}{s^2+4s+25} \end{bmatrix} \quad (10)$$

$$= \begin{bmatrix} \frac{s^2+5s+29}{s^2+4s+25} & \frac{s+5}{s^2+4s+25} \\ -\frac{25}{s^2+4s+25} & \frac{s^2+4s}{s^2+4s+25} \end{bmatrix} = \left(\frac{1}{s^2 + 4s + 25} \right) \begin{bmatrix} s^2 + 5s + 29 & s + 5 \\ -25 & s^2 + 4s \end{bmatrix} \quad (11)$$

2 Problem 2

Consider the feedback control system below:

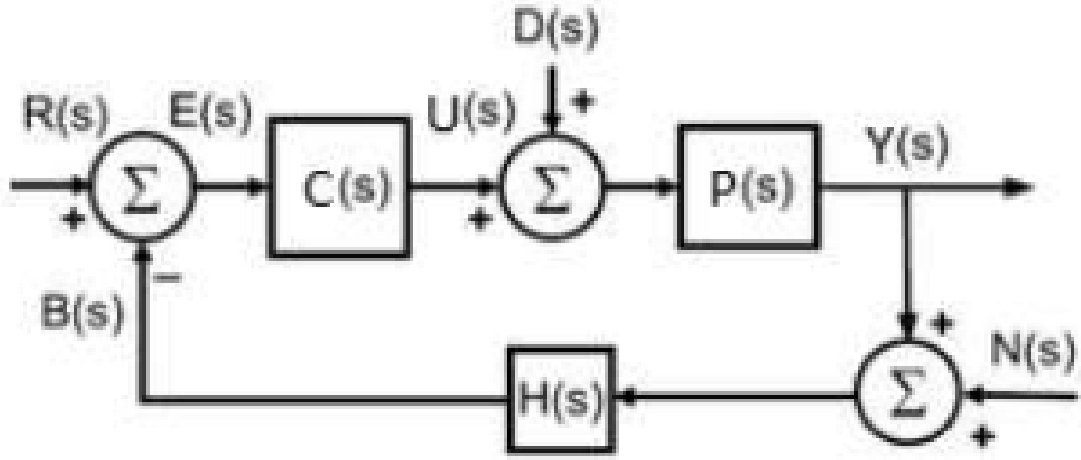


Figure 1: Feedback Control System for Problem 2

2.a

Per the block diagram, define the following Laplace space functions:

$$E(s) = R(s) - B(s) \quad (12)$$

$$U(s) = C(s)E(s) = C(s)(R(s) - B(s)) \quad (13)$$

$$U'(s) = U(s) + D(s) = C(s)E(s) + D(s) \quad (14)$$

$$Y(s) = P(s)U'(s) = P(s)(C(s)E(s) + D(s)) \quad (15)$$

$$B(s) = H(s)(Y(s) + N(s)) \quad (16)$$

where U' is the function directly to the left of the $P(s)$ block.

Therefore, we can do a long sequence of algebraic manipulations to find the requested transfer functions:

$$Y = P(CE + D) \quad (17)$$

$$E = R - B = R - H(Y + N) \quad (18)$$

$$\Rightarrow (Y = P[C(R - H(Y + N)) + D]) \quad (19)$$

$$Y = PCR - PCHY - PCHN + PD \quad (20)$$

$$\Rightarrow (Y(1 + PCH) = PCR - PCHN + PD) \quad (21)$$

$$\Rightarrow \left(Y = \frac{PCR - PCHN + PD}{1 + PCH} \right) \quad (22)$$

$$\Rightarrow \left(Y = \left(\frac{PC}{1 + PCH} \right) R - \left(\frac{PCH}{1 + PCH} \right) N + \left(\frac{P}{1 + PCH} \right) D \right) \quad (23)$$

$$\left(\frac{Y(s)}{R(s)} = \frac{PC}{1 + PCH} \right), \quad \left(\frac{Y(s)}{N(s)} = - \left(\frac{PCH}{1 + PCH} \right) \right), \quad \left(\frac{Y(s)}{D(s)} = \left(\frac{P}{1 + PCH} \right) \right) \quad (24)$$

We can also consider E directly to find the remaining two transfer functions, when paired with extra constraints:

Let $((N = 0) \wedge (D = 0))$:

$$E = R - H(Y + N) = R - HY \quad (25)$$

$$\frac{E(s)}{R(s)} = 1 - H \left(\frac{Y}{R} \right) = 1 - H \left(\frac{PC}{1 + PCH} \right) = \frac{1}{1 + PCH} \quad (26)$$

Let $((N = 0) \wedge (R = 0))$:

$$E = R - H(Y + N) = -HY \quad (27)$$

$$\frac{E(s)}{D(s)} = -H \left(\frac{Y}{D} \right) = -H \left(\frac{P}{1 + PCH} \right) = -\frac{HP}{1 + PCH} \quad (28)$$

2.b

Let $(P(s) = K/\tau s + 1)$, $(C(s) = K_p)$, and $(H(s) = 1)$.

2.b.i

Recall:

$$\frac{E(s)}{R(s)} = \frac{1}{1 + PCH} \quad (29)$$

We can rewrite this transfer function as follows:

$$\frac{E(s)}{R(s)} = \frac{1}{1 + \left(\frac{KK_p}{\tau s + 1} \right)} = \frac{\tau s + 1}{\tau s + 1 + KK_p} \quad (30)$$

If $R(s) = \frac{1}{s}$ (unit step input):

$$E(s) = \frac{\tau s + 1}{s(\tau s + 1 + KK_p)} \quad (31)$$

Therefore, the steady state error response for a unit step reference input $e_{ss, R}$ is (by FVT):

$$e_{ss, R} = \left(\lim_{s \rightarrow 0} \right) (sE(s)) = \left(\lim_{s \rightarrow 0} \right) \left(\frac{\tau s + 1}{\tau s + 1 + KK_p} \right) = \frac{1}{1 + KK_p} \quad (32)$$

2.b.ii

Recall:

$$\frac{E(s)}{D(s)} = -\frac{HP}{1 + PCH} \quad (33)$$

We can rewrite this transfer function as follows:

$$\left(\frac{E(s)}{D(s)}\right) = -\left(\frac{K/\tau s+1}{1 + (KK_p/\tau s+1)}\right) = -\left(\frac{K}{\tau s + 1 + KK_p}\right) \quad (34)$$

If $D(s) = \frac{1}{s}$ (unit step disturbance input):

$$E(s) = -\left(\frac{K}{\tau s + 1 + KK_p}\right)\left(\frac{1}{s}\right) \quad (35)$$

Therefore, the steady state error response for a unit step disturbance input $e_{ss, D}$ is (by FVT):

$$e_{ss, D} = \left(\lim_{s \rightarrow 0}\right)(sE(s)) = \left(\lim_{s \rightarrow 0}\right)\left(-\frac{K}{\tau s + 1 + KK_p}\right) = -\frac{K}{1 + KK_p} \quad (36)$$

2.b.iii

Recall:

$$(E = -HY - HN) \quad (37)$$

$$\Rightarrow \left(\frac{E}{N} = -H\left(\frac{Y}{N}\right) - H = \frac{PCH^2}{1 + PCH} - H = \frac{PCH^2 - (H + PCH^2)}{1 + PCH}\right) \quad (38)$$

$$= -\frac{H}{1 + PCH} \quad (39)$$

We can rewrite this transfer function as follows:

$$\left(\frac{E(s)}{N(s)}\right) = -\left(\frac{1}{1 + (KK_p/\tau s+1)}\right) = -\frac{\tau s + 1}{\tau s + 1 + KK_p} = \frac{E(s)}{R(s)} \quad (40)$$

Therefore,

$$e_{ss, N} = e_{ss, R} = \frac{1}{1 + KK_p} \quad (41)$$

3 Problem 3

Consider the feedback control system below:

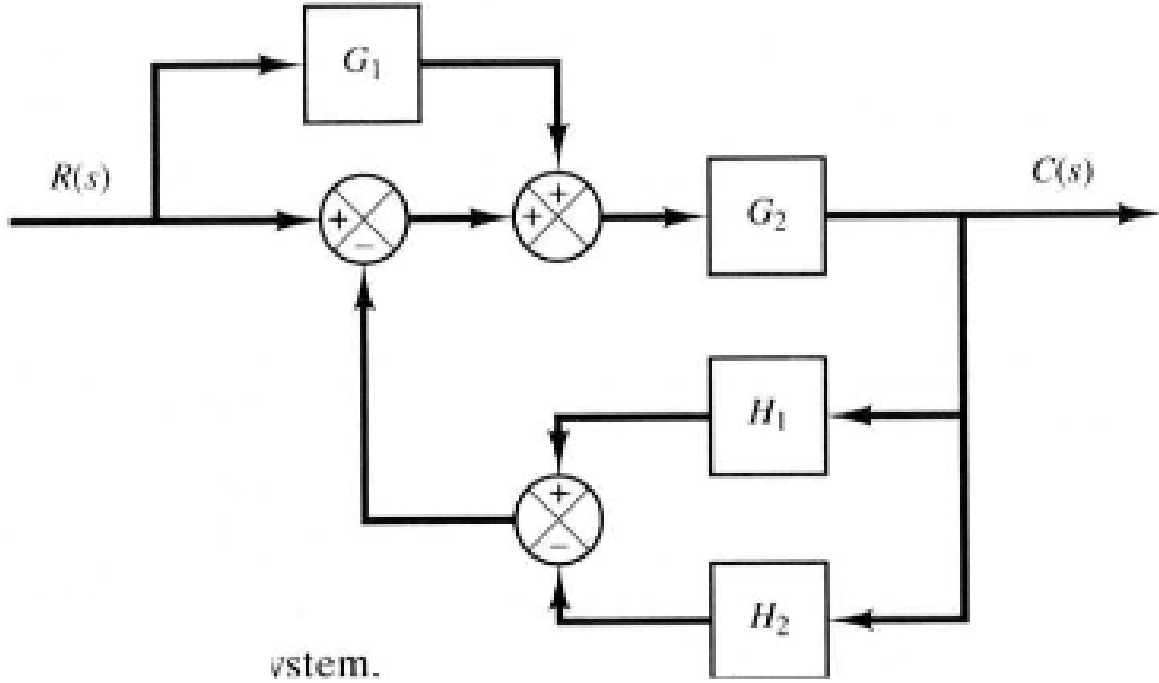


Figure 2: Feedback Control System for Problem 3

Firstly, notice the H_1 and H_2 blocks are in parallel relative to a summing block, so you can replace those two blocks as well as the summing block with $(H_1 - H_2)$. Now, consider the two summing blocks to the right of G_2 , labelled Σ_1 and Σ_2 from left to right. Considering this block diagram, we can express the equations that follow each respective summing block in the following way:

$$\Sigma_1 = R - C(H_1 - H_2) \quad (42)$$

$$\Sigma_2 = \Sigma_1 + G_1 R = R - C(H_1 - H_2) + G_1 R = R(1 + G_1) - C(H_1 - H_2) \quad (43)$$

However, per the block diagram, ($C = G_2 \Sigma_2$), therefore we can begin a chain of simplifications to find the final relevant transfer function $\left(\frac{C(s)}{R(s)} \right)$:

$$C = G_2 \Sigma_2 = G_2 (R(1 + G_1) - C(H_1 - H_2)) = G_2 R(1 + G_1) - G_2 C(H_1 - H_2) \quad (44)$$

$$\Rightarrow (C + G_2 C(H_1 - H_2) = C(1 + G_2(H_1 - H_2)) = G_2 R(1 + G_1)) \quad (45)$$

$$\frac{C(s)}{R(s)} = \frac{G_2(1 + G_1)}{1 + G_2(H_1 - H_2)} \quad (46)$$

4 Problem 4

Consider the feedback control system below:

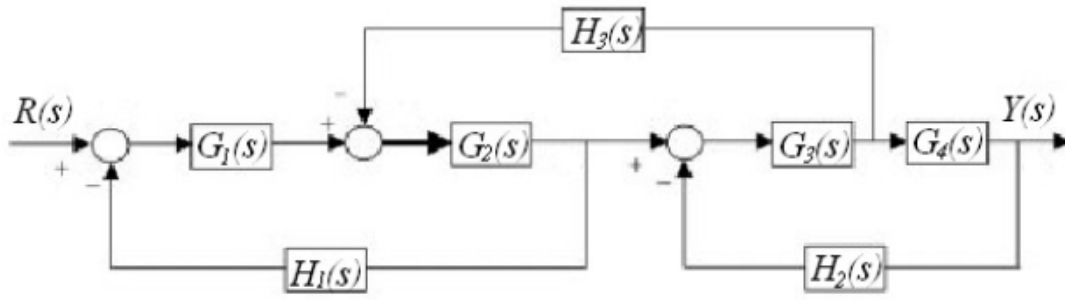


Figure 3: Feedback Control System for Problem 4

Firstly, you can move the summing point of the negative feedback loop containing H_3 outside of the positive feedback loop containing G_1 , requiring that we place $\left(\frac{H_3}{G_1}\right)$ in the block instead, and move the summing block to the intersection between G_4 , Y , and H_2 .

Then, G_1 and G_2 are in series with one another, so we can replace both blocks with one block containing $G_1 G_2$.

Additionally, you can move the origin of the negative feedback loop containing $\frac{H_3}{G_1}$ the right of the block containing G_4 , requiring that we place $\left(\frac{H_3}{G_1 G_4}\right)$ in the block instead.

Then, G_3 and G_4 are in series with one another, so we can replace both blocks with one block containing $G_3 G_4$.

We can now remove the loop containing $G_3 G_4$ and H_2 , replacing it with a block containing $\left(\frac{G_3 G_4}{1 + G_3 G_4 H_2}\right)$.

You can move the origin of the negative feedback loop containing H_1 the right of the block containing $\left(\frac{G_3 G_4}{1 + G_3 G_4 H_2}\right)$, requiring that we place $\left(\frac{H_1(1 + G_3 G_4 H_2)}{G_3 G_4}\right)$ in the block instead.

Now, the $G_1 G_2$ and the $\left(\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2}\right)$ are in series, so we can replace both blocks with one block containing:

$$\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2} \quad (47)$$

We can now remove the loop containing $\left(\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2}\right)$ and $\left(\frac{H_1(1 + G_3 G_4 H_2)}{G_3 G_4}\right)$, replacing it with a block containing:

$$\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2} \left(1 + \left(\frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2}\right) \left(\frac{H_1(1 + G_3 G_4 H_2)}{G_3 G_4}\right)\right)^{-1} \quad (48)$$

$$= \frac{G_1 G_2 G_3 G_4}{1 + G_3 G_4 H_2} (1 + G_1 G_2 H_1)^{-1} \quad (49)$$

$$= \frac{G_1 G_2 G_3 G_4}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)} \quad (50)$$

Finally, we can remove the loop containing $\left(\frac{G_1 G_2 G_3 G_4}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)}\right)$ and $\left(\frac{H_3}{G_1 G_4}\right)$ to get the transfer function of this control system:

$$= \frac{G_1 G_2 G_3 G_4}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)} \left(1 + \left(\frac{G_1 G_2 G_3 G_4}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)} \right) \left(\frac{H_3}{G_1 G_4} \right) \right)^{-1} \quad (51)$$

$$= \frac{G_1 G_2 G_3 G_4}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)} \left(1 + \left(\frac{G_2 G_3 H_3}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)} \right) \right)^{-1} \quad (52)$$

$$= \frac{G_1 G_2 G_3 G_4}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)} \left(\left(\frac{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1) + G_2 G_3 H_3}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)} \right) \right)^{-1} \quad (53)$$

$$= \frac{G_1 G_2 G_3 G_4 (1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1)((1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1) + G_2 G_3 H_3)} \quad (54)$$

$\frac{Y(s)}{R(s)} = \frac{G_1 G_2 G_3 G_4}{(1 + G_3 G_4 H_2)(1 + G_1 G_2 H_1) + G_2 G_3 H_3} \quad (55)$
